

HITESH YADAV

New Delhi

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EDUCATION

Thapar Institute of Engineering and Technology, Patiala **2023 – 2027**
B.Tech - Computer Science and Engineering - CGPA - 9.82 *Patiala, Punjab*

Shanti Gyan Niketan Sr. Sec. School , Delhi **2021 – 2023**
CBSE (Class XII – 2023)- 95% *New Delhi*

Experience

Algo University x Tensor School of CS & AI (YC S21) **June 2026**
Tech Intern

- Architected a 7-container distributed microservices backend using Django, Redis, and Celery, implementing direct-to-S3 multipart uploads that successfully processed 10GB+ video files and reduced server memory overhead by 85%.
- Engineered a high-concurrency write-behind caching system leveraging Redis Hashes to absorb 5,000+ continuous video progress heartbeats per minute, reducing PostgreSQL database write load by 95% via Celery Beat bulk flushes.
- Applied strict anti-piracy controls for 10,000+ concurrent viewing sessions using AES-128 encrypted HLS streaming and O(log N) Redis ZSET locks, ensuring zero-latency session eviction and secure content delivery.

Unessa Foundation **January 2026**
Data Science Intern

- Analyzed social datasets of 1k+ records, identifying trends that improved program insights by 20–25%.
- Built dashboards and analytical reports using Power BI, reducing manual data analysis time by 40% and improving reporting accuracy by 25%.
- Contributed to Foundation's social mission by supporting community empowerment initiatives, analyzing datasets across 3+ programs to enable evidence-based decision-making and improve outreach effectiveness.

PROJECTS

Pacman AI Game  | [Pygame, Pathfinding, BFS, A*, Hill-Climbing, Grid-Based Navigation](#) **Nov 2024**

- Built a 2D Pacman game using Pygame with smooth animations, collision detection, and scoring system.
- Executed AI-controlled ghost navigation using BFS, A*, and Hill-Climbing, achieving 35–45% faster path convergence compared to random movement and 100% guaranteed shortest-path in BFS-based scenarios.
- Designed a responsive, event-driven GUI supporting 60+ FPS real-time interactions, with 16 ms frame update latency, ensuring smooth animations and consistent gameplay behavior across all game states.

Road Sign Detection  | [Python, YOLOv8, OpenCV, Ultralytics, Roboflow, PyTorch](#) **Jun 2025**

- Trained a custom YOLOv8 model on a Roboflow traffic dataset consisting of 3530 training, 801 validation, and 641 test images.
- Achieved mAP@0.5 81 %, Recall 96%, and Precision 60%, indicating strong object coverage with some false-positive detections.
- Analyzed model performance using confusion matrices and precision–recall curves, identifying that 65–70% of false positives arose from visually similar road sign classes, such as speed-limit and warning signs.

StudyNotion –EdTech Platform  | [MongoDB, ExpressJS, React, NodeJS, Redux, Razorpay](#) **Jan 2025**

- Built a full-stack EdTech platform with JWT-based authentication and role-based authorization supporting 3 user roles (admin, instructor, student).
- Developed 15+ RESTful APIs in Express.js for course management, user profiles, enrollment, and progress tracking using MongoDB schemas.
- Implemented Redux-based global state management across 10+ application pages, ensuring consistent user flow between authentication, dashboards, and course modules.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML, CSS, MATLAB, R, NoSQL, SQL

Frameworks: React.js, Express.js, Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, OpenCV

Developer Tools: Git, GitHub, VS Code, Docker, POSTMAN, Node.js, IBM Composer

ACHIEVEMENTS

- Kaggle Datasets Expert with a global rank of 190 out of 8,427 contributors, placing in the top 2.2%.
- Kaggle Notebooks Expert with a global rank of 1,656 out of 58,851 contributors, placing in the top 2.8%.
- Awarded merit-based scholarships worth 14+ Lakhs by Shanti Gyan Niketan School and Thapar University.
- Solved 800+ problems on platforms like GFG , LeetCode,Hackerrank demonstrating strong coding skills.
- Winner of Hacklipse 4.0, an intra-college hackathon organized by the ACM Computing Chapter.
- 1st Runner-Up of Kaggle Hackathon organized by Thapar Institute of Engineering and Technology.